

ISOMETRIC DRAWING AUTO-EDIT

E3D-SA Isometric Edit

Cleanup for E3D Stress (SA) isometrics — 11 steps following SA output rules (delete tee text / spec-no / flange info, etc.).

Key Purpose

Why this program

1. One-click cleanup of E3D Stress (SA) isometrics

– Chains 12 edit steps (sae3d_00~11) in memory on each DXF and saves one _FINAL.dxf

→ **Effect:** A whole batch of stress ISOs is tidied in seconds with no manual step-by-step editing or intermediate files

2. Delete clutter text per SA output rules

– Removes small bore-size callouts (DN ≤ chosen threshold), tee text, spec-no and flange info

→ **Effect:** Drawings stay readable — only the sizes that matter remain, exactly to the SA drawing standard

3. Normalise elevation prefix & fonts

– EL keeps as-is, or FL/GL replaces every 'EL+' label; one font applied to all drawing text

→ **Effect:** Consistent elevation notation and typography across every sheet with zero hand-editing

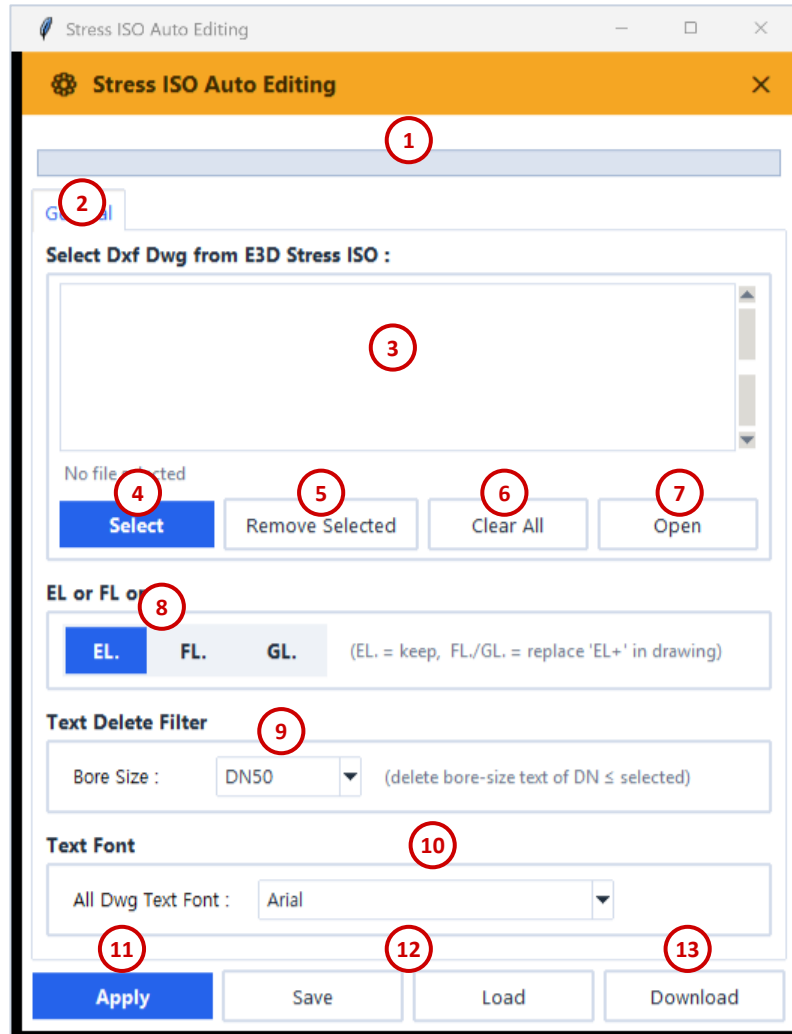
4. Runs fully offline in one dialog

– Select drawings, set options, Apply — steps' pop-ups are auto-suppressed and batched

→ **Effect:** No clicking through each tool's dialogs; options can be saved/loaded and reused per project

Required inputs : One or more E3D Stress ISO DXF drawings

Main Window – Item Guide



1. Progress bar (batch status)
2. General tab
3. Selected DXF drawings list
4. Select DXF drawings
5. Remove Selected
6. Clear All
7. Open selected drawing
8. EL / FL / GL elevation prefix
9. Bore Size text-delete filter (DN ≤ selected)
10. All Dwg Text Font
11. Apply — run cleanup on drawings
12. Save / Load options
13. Download (copy drawings to folder)

Processing Steps

What 'Apply' runs, in order

1. Convert polyline outlines to clean single lines
2. Rename pipe layers by line number and size
3. Clean up and reformat dimension lines
4. Redraw weld symbols
5. Edit slope (fall) markers
6. Remove clutter entities around branch olets
7. Delete flange information text
8. Reformat tee-branch text into boxes
9. Remove instrument callout markers
10. Add support tags and node numbers
11. Delete spec-number text
12. Final finishing pass
13. Replace elevation prefix (EL to FL/GL) where selected
14. Delete small bore-size callout text (DN below threshold) and save as _FINAL.dxf

Overview

What it does

1. Cleanup for E3D Stress (SA) isometrics — 11 steps following SA output rules (delete tee text
2. Spec-no
3. Flange info
4. Etc.)

Category: Isometric Drawing Auto-Edit

Folder: ISO_AUTO_EDIT/ISO_AUTO_EDIT_FOR_E3DSA

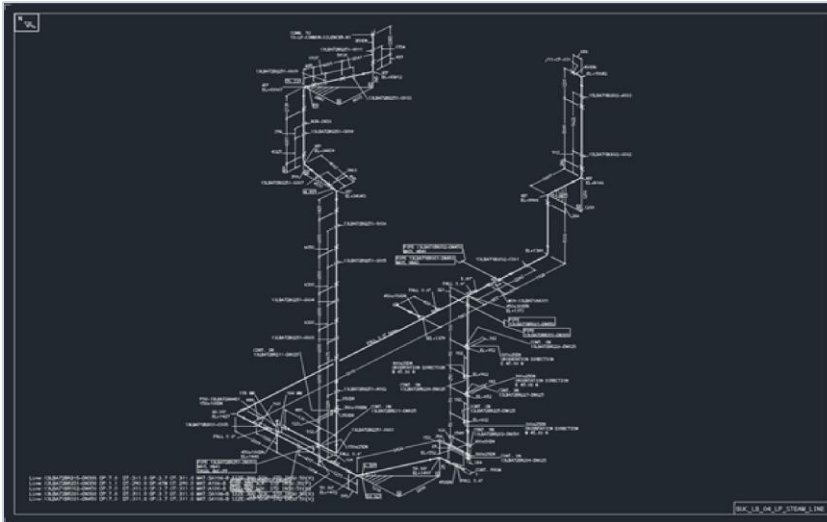
Main scripts: SAE3D_ON_RUNTEST.PY

Scripts / Status: 27 scripts · Operational

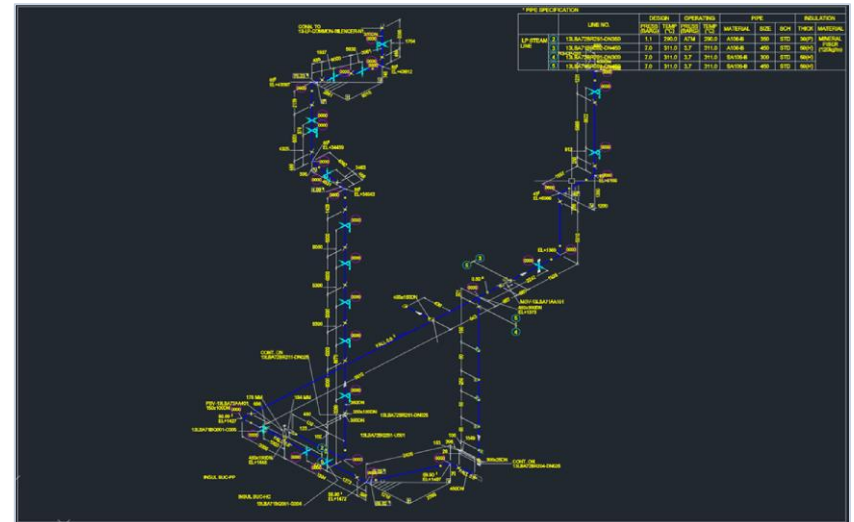
Tags: E3D-SA · TEE · SPEC

Output Samples

Output Samples



E3D-SA ISO Input File



E3D-SA ISO Auto Output File

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Isometric Drawing Auto-Edit.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- Launch from the PC launcher (런처.bat) or run the main script: SAE3D_ON_RUNTEST.PY.
- Output samples and the program window above reflect the current build.

Thank You

E3D-SA Isometric Edit · Plant Engineering Total Service System